

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/305784636>

The Oxford Happiness Questionnaire: Transformation from an Ordinal to an Interval Measure Using Rasch Analysis

Article in *Journal of Happiness Studies* · October 2017

DOI: 10.1007/s10902-016-9784-3

CITATIONS

49

READS

34,087

6 authors, including:



Oleg N. Medvedev

University of Waikato

217 PUBLICATIONS 3,142 CITATIONS

[SEE PROFILE](#)



Richard J Siegert

Auckland University of Technology

304 PUBLICATIONS 10,926 CITATIONS

[SEE PROFILE](#)



Ahmed D Mohamed

MidCentral District Health Board

19 PUBLICATIONS 283 CITATIONS

[SEE PROFILE](#)



Daniel Shepherd

Auckland University of Technology

163 PUBLICATIONS 3,555 CITATIONS

[SEE PROFILE](#)

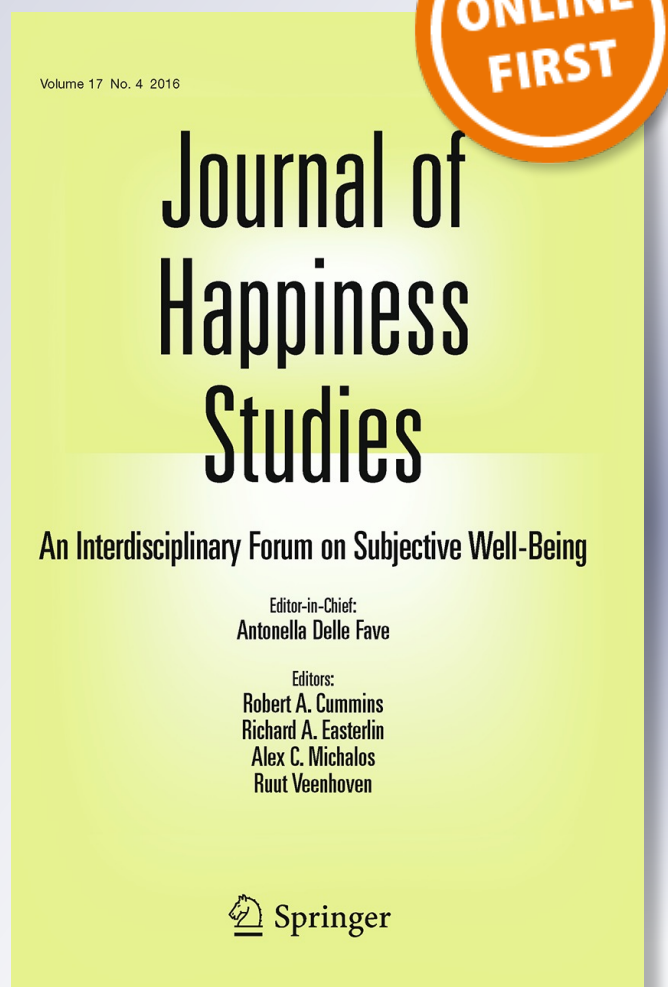
The Oxford Happiness Questionnaire: Transformation from an Ordinal to an Interval Measure Using Rasch Analysis

**Oleg N. Medvedev, Richard J. Siegert,
Ahmed D. Mohamed, Daniel Shepherd,
Erik Landhuis & Christian U. Krägeloh**

Journal of Happiness Studies
An Interdisciplinary Forum on
Subjective Well-Being

ISSN 1389-4978

J Happiness Stud
DOI 10.1007/s10902-016-9784-3



Your article is protected by copyright and all rights are held exclusively by Springer Science +Business Media Dordrecht. This e-offprint is for personal use only and shall not be self-archived in electronic repositories. If you wish to self-archive your article, please use the accepted manuscript version for posting on your own website. You may further deposit the accepted manuscript version in any repository, provided it is only made publicly available 12 months after official publication or later and provided acknowledgement is given to the original source of publication and a link is inserted to the published article on Springer's website. The link must be accompanied by the following text: "The final publication is available at link.springer.com".

The Oxford Happiness Questionnaire: Transformation from an Ordinal to an Interval Measure Using Rasch Analysis

Oleg N. Medvedev^{1,5} · Richard J. Siegert¹ · Ahmed D. Mohamed^{2,3} · Daniel Shepherd¹ · Erik Landhuis⁴ · Christian U. Krägeloh¹

© Springer Science+Business Media Dordrecht 2016

Abstract The scientific study of happiness requires accurate measurement of the construct that satisfies assumptions of parametric statistics and thus allows both researchers and clinicians to make reliable and valid comparisons with the relevant data sources. The 29-item Oxford Happiness Questionnaire (OHQ) is a widely-used scale for assessment of personal happiness. While its psychometric properties are acknowledged to be acceptable, it presents scores on an ordinal scale and may thus not discriminate precisely between individual happiness levels. The current study aimed to improve precision and item functioning of the OHQ by applying Rasch analysis to a sample of 281 participants. To

✉ Oleg N. Medvedev
oleg.medvedev@aut.ac.nz

Richard J. Siegert
richard.siegert@aut.ac.nz

Ahmed D. Mohamed
ahmed.mohamed@cantab.net

Daniel Shepherd
dashephe@aut.ac.nz

Erik Landhuis
clandhui@aut.ac.nz

Christian U. Krägeloh
chris.krageloh@aut.ac.nz

¹ School of Public Health and Psychosocial Studies, Auckland University of Technology, Auckland, New Zealand

² Department of Psychology, Faculty of Science, University of Nottingham, Malaysia Campus, Jalan Broga, 43500 Semenyih, Selangor Darul Ehsan, Malaysia

³ Bioethics Centre, University of Otago, 71 Frederick St, Dunedin 9016, New Zealand

⁴ School of Social Sciences and Public Policy, Auckland University of Technology, Auckland, New Zealand

⁵ Department of Psychology, Faculty of Health and Environmental Sciences, Auckland University of Technology, North Shore Campus, Private Bag 92006, Auckland 1142, New Zealand

correct disordered thresholds items were rescored in a uniform fashion. Four items displayed poor relationships with the latent trait of happiness and were removed. Best fit to the unidimensional Rasch model was achieved after locally dependent items were combined into subtests and adjusted for personal differences. Using the ordinal-to-interval conversion tables published here, ordinal OHQ scores can now be transformed to interval level data and thus subjected to parametric statistical analysis without violating fundamental assumptions. The precision of the instrument can be improved significantly by these minor modifications without the need to modify the original response format.

Keywords Oxford Happiness Questionnaire · Happiness · Subjective well-being · Measurement · Rasch analysis

1 Introduction

Happiness has been the most important goal for humans throughout history (Compton 2005). Aristotle considered his *eudaimonia*, usually translated as ‘happiness’, to be the ultimate goal of humans, and superior to all other goals (Diener et al. 1998). Eudaemonic happiness was suggested to be related to psychosocial functioning and distinct, albeit correlated, with pleasure-driven hedonic happiness (Joshani 2015). Cross-cultural research conducted in 47 countries indicated that happiness is rated higher than all other personal values such as health, love or wealth (Kim-Prieto et al. 2005). Happiness has also been found to be a highly valued component of life quality, superior to other values such as money, health or sex (Skevington et al. 1997). Therefore, happiness can be considered the most desirable condition among humans, and other goals may only be valued as potential determinants of happiness (Csikszentmihalyi 1992). From an evolutionary perspective, happiness can be seen as a psychological reward for adaptive functioning associated with evolutionary fitness (Nesse 1990).

The early tendency of psychological research was to focus mainly on mental illness and abnormalities associated with social or occupational dysfunction (Argyle 2001; Carr 2004) and related cognitive distortions (Beck 1991; Ellis 2002). Towards the end of the 20th century, psychologists started to display an increased interest in the positive dimensions of human life (e.g. well-being, happiness and quality of life), reflected by the growing body of research focused on these constructs (Argyle 2001; Diener 1984). Happiness and subjective well-being are often used as synonyms (Diener 1984, 2006) and became mainstream in economic research contexts (Kristoffersen 2010). Specifically, subjective well-being data are widely used and studied along with economic indicators (Kahneman and Krueger 2006; Spruk and Kešeljčević 2015). For instance, in the past few years, the Organisation for Economic Co-operation and Development (OECD) has been measuring and reporting the Better Life Index (OECD 2015), indicating that subjective well-being or happiness play a crucial role in the life of individuals living in developed nations. However, scientific study of happiness requires accurate measurement of the construct that satisfies assumptions of parametric statistics and thus allows both researchers and clinicians to make reliable and valid comparisons with the relevant data sources—such as comparing the mean values from different sample groups.

Theoretically, happiness can be explained by bottom-up processes representing the affective component of happiness, and top-down processes, relating to cognitive

components (Andrews and McKennell 1980). According to the bottom-up approach, happiness is evaluated as a total sum of aggregated positive and negative feelings (Diener 1984). It should be noted that accurate measurement should include both positive and negative affect because a number of studies have suggested that positive affect is not the opposite of negative affect (Andrews and McKennell 1980; Argyle 2001; Brandburn 1969), and the correlation between them is only moderate (Tellegen et al. 1988). On the other hand, top-down approaches suggest that happiness is largely a product of an individual's cognitions and refers to subjective evaluations of one's experiences and expressions of life satisfaction (Andrews and McKennell 1980; Diener 1984). Even though the top-down approach has been well argued with some supporting evidence (Andrews and McKennell 1980; Beck 1991; Diener 1984), it has been shown that the subjective evaluation of life events influences emotional responses (LeDoux 2000; Medvedev et al. 2015) meaning that the approaches are likely to complement each other. Accordingly, there is research evidence indicating a single dimension of happiness that includes positive affective, negative affective and cognitive components (Argyle 2001; Hills and Argyle 1998, 2002; Joseph and Lewis 1998). Cognitive components may also include personality traits like internal locus of control, extraversion and optimism (Carr 2004; Fordyce 1988; Mayers 1992).

Different measures have been developed to assess happiness, however, operational definitions of these instruments appear inconsistent (Andrews and McKennell 1980; Argyle 2001; Brandburn 1969). Inconsistency in defining subjective happiness and well-being constructs, and considering their relevant components, has resulted in various limitations of existing happiness measures (Eid and Larsen 2008). The most common limitation is not considering both cognitive and affective facets as suggested by theory and research (Andrews and McKennell 1980; Argyle 2001; Diener 1984; Fordyce 1988).

The 29-item Oxford Happiness Questionnaire (OHQ) (Hills and Argyle 2002) is a widely-used scale to assess personal happiness. It is a new version of the original Oxford Happiness Inventory (OHI) (Argyle et al. 1989), which contains 29 items, each accompanied by four statements representing response options similar to the Beck Depression Inventory (Argyle et al. 1989; Beck et al. 1996). The OHI has been scrutinized using Mokken scaling analysis probing whether questionnaire items can correspond reliably to the range of individual abilities on a latent trait (Stewart et al. 2010). The authors concluded that out of 29 items only 12 items could reliably reflect individual happiness levels.

Hills and Argyle (2002) used terms well-being, subjective well-being, and psychological well-being as synonymous to happiness in describing the OHI and OHQ. Both measures are based on theoretical considerations supported by research findings indicating a single dimension of happiness that covers positive and negative affect, and cognitive evaluations such as life satisfaction and happy traits (Andrews and McKennell 1980; Argyle 2001; Diener 1984). Furthermore, there are items reflecting specific cognitive components and traits found within the single happiness factor labeled as sociability, sense of control, physical fitness, positive cognition, mental alertness, self-esteem, cheerfulness, optimism and empathy (Hills and Argyle 1998, 2002). From a theoretical perspective, the item content of the OHQ was criticized for being too broad—spreading over many areas of human experience rather than specifically focusing on happiness (Kashdan 2004).

Hills and Argyle (2002) validated the 29-item OHQ using a 6-point Likert-scale response format worded as *strongly disagree*, *moderately disagree*, *slightly disagree*, *slightly agree*, *moderately agree*, and *strongly agree* and reported a Cronbach's alpha just above 0.90. A five-point response format worded as *agree strongly*, *agree*, *uncertain*, *disagree*, and *disagree strongly* has also been used for the OHQ, and comparable

Cronbach's alpha of 0.90 were reported (Robbins et al. 2010). Due to the large number of factors extracted using the Kaiser criterion in principal component factor analysis (eigenvalue > 1.00), Hills and Argyle (2002) used Direct Oblimin rotation method, which extracted only one second-order component that was interpreted as evidence of unidimensionality.

The OHQ is an ordinal scale and as such it should not be used with parametric statistics such as ANOVA without violating fundamental assumptions of these tests. An ordinal scale cannot be considered as an interval scale by adding individual item scores together or simply by arguing that it is a widely-used scale (Stucki et al. 1996; Allen and Yen 1979). Usage of the ordinal OHQ in research may compromise the validity of comparisons with neurophysiological data (e.g. electroencephalogram, skin conductance, heart rate), an especially important consideration in the modern wellbeing research. Therefore, further research is necessary to improve the precision of the OHQ up to an interval-level measure, and in particular Rasch analysis, a method that is specifically suited for this purpose (Medvedev et al. 2016a; Tennant and Conaghan 2007). Rasch analysis employs a probabilistic mathematical model based on item response theory and uses advanced strategies to improve functioning of both a full scale and individual items. It provides algorithms that allow researchers and clinicians to transform ordinal scores to interval-level data suitable for parametric statistics (Rasch 1960; Tennant and Conaghan 2007).

Rasch analysis has shown numerous advantages over other more traditional statistical methods, which has been sufficiently argued elsewhere (Rasch 1960; Wilson 2005; Wright and Stone 1979). Rasch modelling estimates the probability that any response category of each individual item is endorsed by considering the item's difficulty (location) and the person's ability on the latent trait being measured (i.e. happiness), whereas non-parametric methods do not distinguish between difficulty and ability (Fox and Jones 1998). The analysis investigates several important psychometric parameters such as response option ordering in polytomous items, correct stochastic ordering of items, potential item bias and assumptions of local independence (e.g. unidimensionality) and structural validity (Tennant and Conaghan 2007). When these requirements are satisfied and data fit to the Rasch model, both the items and the participants are ordered on an interval scale measuring the latent trait (e.g. happiness) using the same log-odds metric (Pomplun and Custer 2005). Therefore, an item-person threshold distribution can be represented graphically, showing how well the range of the items' *difficulty* covers the sample's *ability*. Rasch analysis provides a systematic method that can precisely and rigorously diagnose the scale and individual items in terms of potential item bias or differential item functioning (DIF), and unidimensionality. The end product of Rasch analysis is algorithms to transform scores from an ordinal to an interval scale that increase precision of measurement, which was demonstrated empirically (Norquist et al. 2004).

Taken together, the OHQ (Hills and Argyle 2002) has generally accepted psychometric properties, but its item functioning and ability to discriminate precisely between levels of subjective wellbeing has not been sufficiently investigated. Rasch analysis is a suitable technique to test structural validity and to improve the ability of an instrument and individual items to discriminate on their overarching latent trait. However, to the best of our knowledge, Rasch analysis has not yet been used to investigate and to advance the psychometric properties of the OHQ. The present study aimed to apply modern Rasch strategies to improve the psychometric properties of the OHQ up to an interval-level scale. Thus, to enable researchers to analyse the OHQ data using parametric statistics and to make valid comparisons with neurophysiological data (e.g. electroencephalogram, skin conductance, heart rate) ordinal-to-interval conversion tables were produced.

2 Method

2.1 Participants

The sample was based on the recommended optimal sample size estimates for most purposes of Rasch analysis (Linacre 1994) and included 281 university students from two countries: 180 from Auckland University of Technology, New Zealand, and 101 from the University of Nottingham, Malaysia Campus. The New Zealand sample includes 35 males (20 %), 141 females (80 %) and 4 participants who did not provide gender information, and the Malaysian sample had 48 (48 %) males and 53 (52 %) females. The mean age for the pooled sample was 23.70 years, with a standard deviation of 7.32. Ethnicities include 86 (31 %) Caucasian, 36 (13 %) Polynesian, 105 (37 %) Asian, and 45 (16 %) of other ethnic groups. Personal factors such as sample category, gender, ethnic group and age were used to investigate DIFs in Rasch analysis. Three age categories, each representing approximately a third of the age distribution in the sample, were created as follows: 16 to 18 years, 21 to 23 years and 24 to 64 years. Less than 1 percent of the data was missing.

2.2 Instruments

The 29-item OHQ (Hills and Argyle 2002) is a self-report questionnaire that employs a 6-point Likert scale response format from *strongly disagree* = 1 to *strongly agree* = 6, with the higher scores corresponding to higher levels of happiness. There are 12 negatively worded items that require reverse coding before calculating the total happiness score, which is a sum of individual item scores. Examples of positively worded items include 'I am very happy' and negatively worded items 'I rarely wake up feeling rested'. According to Hills and Argyle (2002), the OHQ is supposed to measure personal happiness as a broader unidimensional construct and has good internal consistency, with Cronbach's alpha at the level of 0.90 and above.

2.3 Procedure

The study questionnaires were completed by the New Zealand participants in lecture theaters of Auckland University of Technology before the start of lectures. The data from the Malaysian sample were obtained before the start of a lecture at the University of Nottingham Malaysia campus and as part of a mindfulness and subjective wellbeing neuroscience research study. The Malaysian sample had a good standard of English and scored above 6 points on the International English Language Testing System or 79 points on the Test of English as a Foreign Language (IBT) as part of their entry requirements for their studies. Both studies complied with local and international ethical guidelines and were approved by the ethics committees of the Department of Psychology, Faculty of Health and Environmental Sciences, Auckland University of Technology and the School of Psychology, Faculty of Science, University of Nottingham.

2.4 Data Analysis

Descriptive statistics, internal reliability, and exploratory PCA of the 29-item OHQ were computed using IBM SPSS v.22. Data were then formatted and saved as an ASCII file and imported into the RUMM2030 software for Rasch analysis (Andrich et al. 2009). A

likelihood-ratio test was performed on the initial output analysis and indicated the appropriateness of the unrestricted (Partial-Credit) version of the Rasch model for the current dataset. The sequential stages involved in the Rasch analysis were as follows (Siegert et al. 2010):

1. Testing overall data fit to the expectations of the Rasch model.
2. Screening all 29 OHQ items for disordered thresholds.
3. Examining individual item fit to the Rasch model.
4. Investigating local dependency between 29 OHQ items.
5. Implementing necessary modifications (e.g. removing a misfitting item, creating subtests to address local dependency among items).
6. Re-examining individual item and overall fit to the Rasch model.
7. Testing DIF for gender, age, ethnicity and sample (Malaysia vs New Zealand).
8. Unidimensionality test.
9. Examining the person-item thresholds distribution.
10. Comparing the final solution with the original 29-item OHQ.
11. Comparing Rasch results with traditional psychometric criteria (e.g. item-to-total correlations and loadings on the first principal component).

Rasch analysis is an iterative procedure that starts by testing the overall model fit examining the *threshold map* output to determine whether any items display disordered thresholds. A threshold is disordered if persons' higher levels of a latent factor (e.g. happiness) are not consistently corresponding to choosing progressively higher response options on the ordinal scale for that specific item. Improvement of disordered thresholds can be achieved by collapsing adjacent pairs of response options. Items with disordered thresholds are rescored one or two at a time, followed by re-assessing goodness of fit to the model. If a relatively large number of items displays disordered thresholds, uniform rescoring of all items can be considered (Medvedev et al. 2016a).

After thresholds are appropriately ordered, poorly fitting items are deleted one at a time followed by testing of the overall Rasch model fit. It should be noted that both the individual item and the overall model fit may be affected by local dependency between items. Therefore, thorough examination of the residual correlations matrix is necessary to identify locally dependent items. Items are considered locally dependent if a residual correlation between items is above a conservative cut-off point of 0.20 above the mean of all residual correlations (Marais and Andrich 2008). Issues associated with local dependency can be resolved by combining locally dependent items into a subtest (Wainer and Kiely 1987), and the overall model fit is re-tested once again. This iterative analysis will be repeated until both individual item and overall fit to the Rasch model reach acceptable levels and scale unidimensionality is clearly evident. Item-trait interaction for the overall model fit indexed by Chi-square statistics should be non-significant ($p > 0.05$, Bonferroni adjusted). Overall fit residuals for both person and item are required to have a standard deviation close to 1.00 and a mean close to 0.00 in case of an excellent fit. Fit residuals of individual items are expected to be within the range from -2.50 to $+2.50$. When DIF is examined, there should be no systematic differences in item functioning (Bonferroni adjusted) between personal factors such as gender, age, ethnicity and sample population.

The fundamental assumption of the Rasch model is scale unidimensionality (Rasch 1960, 1961), which is analyzed in RUMM2030 using a unidimensionality test (Smith 2002), explained in greater detail elsewhere (Medvedev et al. 2016b). Person-locations for two groups of items with the highest positive and the highest negative loadings on the first

principal component of the residuals will be compared by an independent-samples t test after the Rasch factor (i.e. happiness) is removed. Unidimensionality of the scale is confirmed if significant t -test comparisons outside the -1.96 to 1.96 confidence intervals do not exceed 5 %. A binominal test for proportions based on sample size can be used to accurately estimate the acceptable deviations from the percentage of significant t -tests. Unidimensionality is evident if the lower bound of a binominal confidence interval computed for the number of significant t -tests overlaps 5 % cut-off point. To examine if there is any significant difference to participant estimates of the modified version, participant estimates for the original scale and the modified version are then calibrated to the same metric and compared using a paired-samples t -test. In Rasch analysis reliability is assessed by the person separation index (PSI), which estimates the scale's ability to discriminate between groups at different trait levels. PSI is numerically close to Cronbach's alpha and similarly estimates the error-free proportion of variance of person estimates distribution relative to the total variance including error. However, its calculation is based on non-linear transformation of raw scores, which can be conducted with random missing data, which is another advantage of Rasch analysis.

3 Results

Internal consistency of the 29-item OHQ with the current dataset was high (Cronbach's alpha 0.89), consistent with the original study (Hills and Argyle 2002). However, five items (2, 5, 7, 14 and 23) displayed low item-to-total correlations (<0.30), and removing any of these items did not result in any noticeable improvement of Cronbach's alpha.

Though not a principal aim of this study, a significantly higher mean happiness level was found for Malaysian students (128.78) compared to New Zealand students (121.20), Mann-Whitney U test, $U = 6371.50$, $p = 01$.

3.1 Initial Test of the Overall Rasch Model Fit

Initial analysis indicated good reliability of the original scale (PSI = 0.90), but the overall fit to the Rasch model was unsatisfactory [$\chi^2(87) = 314.37$, $p < 0.001$, Table 1, Analysis 1]. In Rasch analysis, Chi-square for item-trait interaction should be non-significant meaning that measurement ability is independent from the level of the latent trait possessed by a person. Furthermore, a substantial number of the OHQ items displayed disordered thresholds (items: 1, 3, 6, 7, 9, 10, 11, 16, 23, 24, 27, 28, 29), and threshold ordering of other items was only marginally acceptable. Therefore, items were rescored to correct the thresholds before the analysis continued.

3.2 Rescoring of the OHQ Items

The OHQ items were rescored in an iterative way with subsequent goodness of fit testing. The best fit and optimal ordering of thresholds was achieved when using uniform rescoring of all items by collapsing response categories 'moderately disagree' with 'slightly disagree', and 'slightly agree' with 'moderately agree'. Figure 1 shows disordered thresholds on the response probability curves for item 1 typical for the original OHQ items, and Fig. 2 illustrates the effect of rescoring for the same item, which is illustrative for all other items.

Table 1 Summary of fit statistics for the original OHQ version (1), after uniform item rescoreing (2), after removing items 2, 5, 14 and 23 (3), and the final solution after creating the subtests and splitting items working differently for each sample (4)

Tests	Item residual		Person residual		Goodness of fit		PSI	Independent <i>t</i> -test	
	Value	SD	Value	SD	χ^2 (df)	<i>p</i>		%	LB 95 % CI ^a
1	0.95	2.18	−0.13	1.75	314.37 (87)	<0.001	0.90	28.57	26.02
2	0.31	1.62	−0.30	1.73	222.18 (87)	<0.001	0.89	24.64	22.09
3	0.26	1.59	−0.33	1.69	152.12 (75)	<0.001	0.89	21.79	19.23
4	0.16	0.81	−0.45	0.99	13.03 (15)	0.60	0.82	4.56	1.73

^a LB, lower bound of the 95 % confidence interval

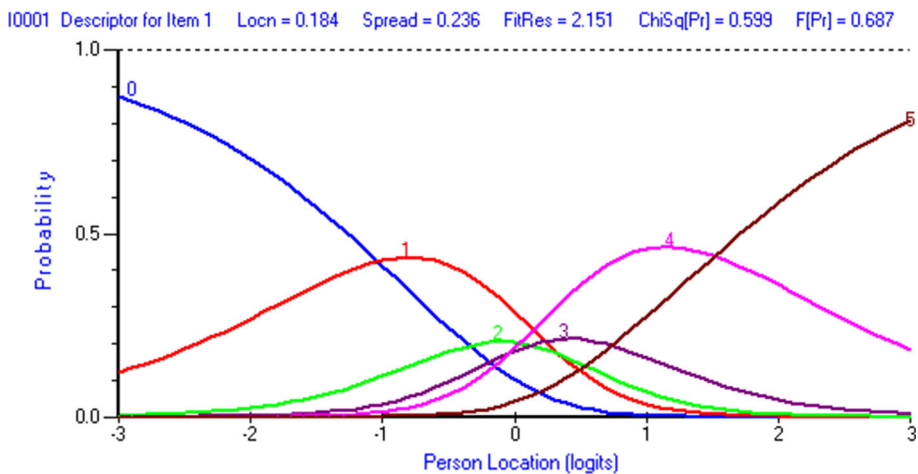


Fig. 1 Response probability curves for item 1 of the OHQ showing disordered thresholds

All disordered thresholds were corrected after uniform rescoreing, and the overall model fit was improved, although it remained below acceptable levels [$\chi^2(87) = 222.19$, $p < 0.001$, Table 1, Analysis 2].

3.3 Removing Items Not Fitting to the Model

After uniform rescoreing, fit residuals of all individual items were analyzed. Table 2 shows item fit residuals and Chi-square values after uniform rescoreing together with item-to-total correlations and loadings on the first principal component for the original OHQ version. Item 5 ‘I rarely wake up feeling rested’ and item 23 ‘I do not find it easy to make decisions’ showed highest fit residuals above 3.00 indicating a poor fit to the Rasch model. Item-to-total correlations and loadings on the first principal component for these items were below 0.30 suggesting poor relationships with the latent trait (Table 2). Both items were removed, with subsequent testing of the overall model fit. Noticeable improvement was evident but expectations of the model were still not reached [$\chi^2(81) = 192.79$, $p < 0.001$, PSI = 0.89].

Further examination of individual items fit residuals indicated that items 2 ‘I am intensely interested in other people’ and 14 ‘There is a gap between what I would like to do

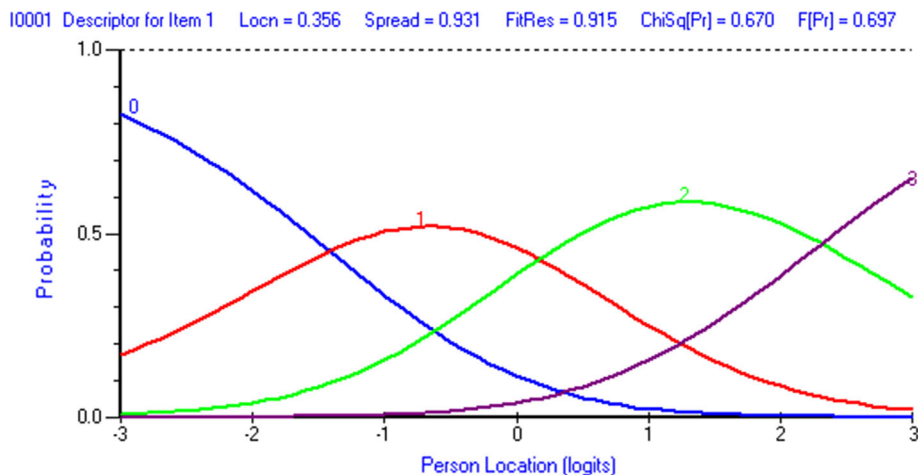


Fig. 2 Response probability curves for item 1 of the OHQ showing ordered thresholds after rescoring

and what I have done' had fit residuals above the 2.50 cutoff point and the highest Chi-square values (above 17) for item-trait interaction, which indicates a poor fit to the Rasch model. Both items also showed low item-to-total correlations and low loadings (<0.30) on the first principal component (Table 2). Removing items 2 and 14 improved the overall model fit, but Chi-square was still at a significant level [$\chi^2(75) = 152.12$, $p < 0.001$, $PSI = 0.89$, Table 1, Analysis 3].

3.4 Local Dependency

At this stage the residual correlation matrix was examined, which revealed three groups of locally dependent items displaying residual correlations above the acceptable level of 0.20 above the mean of all residual correlations (Marais and Andrich 2008). This observation was confirmed by a principal component factor analysis with principal axis factoring using Varimax rotation. Fixing the number of factors to three clearly highlighted the locally dependent items as three distinct groups (Table 3).

One larger group included all negatively worded items (1, 6, 10, 13, 19, 24, 27, 28 and 29); the second group items were more cognitive and thus focusing on attitude (3, 4, 8, 9, 12, 16, 18, 21 and 3); and the third group included mainly items focused on affect (7, 11, 15, 17, 20, 22, 25 and 26). However, items 15, 20, 22, 25 and 26 cross-loaded on both attitude and affect, with consistently higher coefficients representing the affective factor. These three factors correlate at the level of 0.50 and above, providing evidence of an overarching single factor but also confirm local dependency between these three item groups. Therefore, locally dependent items were combined into three subtests providing a desirable alternative to improve fit to the Rasch model without excluding further OHQ items [$\chi^2(9) = 18.00$, $p = 0.35$].

3.5 Differential Item Functioning (DIF)

Testing for DIF examined influence of personal factors such as age, gender, ethnicity, and sample (i.e. New Zealand vs Malaysian students) on functioning of individual items. DIF analysis involves comparing distributions of individual scores aggregated by class intervals

Table 2 Corrected item-to-total correlation and loadings on the first principal component (PC) for the original OHQ, and item-fit residuals and Chi-square values for item-trait interaction after uniform rescoring (1) and after removing items 5 and 23 (2)

Item	Item-to-total correlation	Item loadings on the 1st PC	Item-fit residuals		Item χ^2	
			(1)	(2)	(1)	(2)
1. Pleased with self ^R	0.45	0.44	0.92	1.14	1.55	3.88
2. Interested in others	0.20	0.21	2.52	2.94	15.46	17.96
3. Life is rewarding	0.45	0.49	-0.15	-0.12	0.93	1.00
4. Warmth for others	0.48	0.53	-0.17	-0.20	2.23	2.49
5. Wake up rested^R	0.30	0.28	3.80	-	22.61	-
6. Optimistic ^R	0.47	0.46	0.19	0.60	3.77	6.00
7. Find things amusing	0.28	0.31	1.26	1.36	5.09	6.47
8. Committed and involved	0.41	0.44	0.64	0.89	1.42	4.07
9. Life is good	0.59	0.65	-1.40	-1.47	6.85	7.32
10. World is good ^R	0.35	0.34	2.98	3.40	6.45	10.24
11. Laugh a lot	0.37	0.45	1.27	1.03	0.29	0.09
12. Satisfied with life	0.61	0.69	-1.18	-1.28	11.20	12.96
13. Look attractive ^R	0.50	0.51	-0.35	-0.12	1.03	0.97
14. Done things wanted^R	0.28	0.28	2.36	3.21	9.92	17.56
15. Very happy	0.69	0.76	-2.29	-2.44	19.11	21.53
16. Find beauty in things	0.37	0.42	-0.04	-0.02	2.06	3.80
17. Cheerful effect on others	0.39	0.46	0.12	0.09	1.58	3.15
18. Can organise time	0.47	0.52	0.37	0.61	5.40	6.42
19. Feel in control ^R	0.48	0.46	0.54	1.24	0.18	2.17
20. Feel able do most things	0.57	0.63	-0.97	-1.11	11.00	6.34
21. Mentally alert	0.60	0.64	-1.38	-1.17	4.64	2.48
22. Joy and elation	0.65	0.71	-2.02	-2.13	20.04	16.46
23. Make decisions easily^R	0.23	0.21	3.15	-	27.01	-
24. Life has meaning and purpose ^R	0.48	0.46	0.82	1.42	3.65	1.29
25. Feel energetic	0.59	0.67	-1.46	-1.36	8.42	5.12
26. Good influence	0.58	0.65	-0.99	-1.12	7.80	7.31
27. Have fun with others ^R	0.50	0.48	-1.33	-1.12	11.80	8.10
28. Feel healthy ^R	0.52	0.51	-0.62	-0.21	1.93	1.73
29. Happy memories ^R	0.42	0.41	2.13	2.89	8.78	15.85

^R Negatively worded items. Removed misfitting items 2, 5, 14 and 23 are presented in bold

(CI) mean scores between groups of each personal factor and per each individual item using ANOVA. If the effect of a personal factor (Bonferroni adjusted) is significant for an item, the analysts conducted a visual examination of the item characteristic curve (ICC) with CI means for all groups plotted on the ICC. A significant DIF effect by ethnic group was found for subtests 1 [$F(3, 280) = 5.93, p < 0.001$] and 2 [$F(3, 280) = 5.77, p < 0.001$], Bonferroni adjusted. However, visual examination of plots revealed that these differences were not consistent across observed CIs, with at least one shared CI point at each level, which is considered as non-uniform DIF (Andrich and Hagquist 2013). Also, there was a significant DIF effect by sample for subtest 1 [$F(1, 280) = 59.23, p < 0.001$],

Table 3 Principal component factor analysis with principal axis factoring using Varimax rotation and number of factors fixed to three after misfitting items 2, 5, 14 and 23 were removed ($n = 281$)

Factor items	Negative worded	Cognitive	Affective
27 Have fun with others ^R	0.75		
24 Life has meaning and purpose ^R	0.72		
29 Happy memories ^R	0.70		
1 Pleased with self ^R	0.65		
28 Feel healthy ^R	0.65		
6 Optimistic ^R	0.63		
19 Feel in control ^R	0.59		
10 World is good ^R	0.51		
13 Look attractive ^R	0.46		
8 Committed and involved		0.67	
21 Mentally alert		0.66	
9 Life is good		0.64	
16 Find beauty in things		0.62	
3 Life is rewarding		0.60	
12 Satisfied with life		0.58	
18 Can organise time		0.51	
4 Warmth for others		0.45	
11 Laugh a lot			0.79
17 Cheerful effect on others			0.78
7 Find things amusing			0.56
25 Feel energetic		0.49	0.54
15 Very happy		0.48	0.53
20 Feel able to do most things		0.44	0.48
22 Joy and elation		0.41	0.48
26 Good influence		0.44	0.46

^R Negatively worded items. Coefficients below 0.40 are suppressed for clarity

subtest 2 [$F(1, 280) = 59.68, p < 0.001$] and subtest 3 [$F(1, 280) = 11.73, p < 0.001$], Bonferroni adjusted, but the means were only systematically different for subtests 1 and 2. Therefore, subtests 1 and 2 were split for DIF by sample resulting in the same subtests measuring the New Zealand and the Malaysian sample groups independently. After the subtests were split, DIF analysis was repeated for all personal factors, and no significant DIFs were evident. This solution permitted transformation from ordinal to interval level data for each sample without any further modifications and resulted in the best overall model fit [$\chi^2(15) = 13.03, p = 0.60, \text{PSI} = 0.82$, Table 1, Analysis 4].

3.6 Unidimensionality Test

Unidimensionality of the final model solution was tested by comparing the person estimates from the subset with the highest negative loadings on the first principal component with the estimates from the subset with the highest positive loadings. Strict unidimensionality was evident with the percentage of significant t -tests below 5 % (Table 1).

3.7 Item-Person Thresholds Distribution

Figure 3 illustrates the person-item threshold distribution of the original OHQ after uniform rescaling (Table 1, Analysis 2) and Fig. 4 after removing misfitting items 2, 5, 14 and 23, creating three subtests and splitting subtests by sample DIF (Table 1, Analysis 4). Person ability on the latent factor (happiness) and item difficulty are plotted here using the same metric in logit units. Figure 3 shows extreme misfitting items on the left-hand side that are outside of the sample abilities and, therefore, do not discriminate between individual happiness levels. Also, there is a gap that can be seen between item difficulty levels meaning that not all levels of the latent factor are adequately covered by the original item structure of the OHQ.

However, after the Rasch modifications that involved removing misfitting items, creating subtests and DIF splitting, the person-threshold distribution closely resembles a normal distribution. The modified OHQ item thresholds perfectly cover the abilities of the sample on the latent factor (happiness), with the Malaysian sample showing higher 'abilities' on the latent factor compared to the New Zealand sample (Fig. 4).

3.8 Equating *t*-test

The main purpose of the equating *t*-test is to verify that Rasch modifications produced a significant difference in measuring individual happiness levels by comparing person estimates of the original OHQ and the Rasch modified version using a paired-samples *t*-test. A significant difference was found between the person estimates of the two versions [$t(279) = 21.36, p < 0.01$], which confirmed that the implemented Rasch modifications were successful and resulted in an improved solution for the 25-item OHQ version.

3.9 Ordinal-to-Interval Conversion Tables

Table 4 provides a simple algorithm to convert ordinal OHQ scores to interval-level data. The raw scores presented here should be calculated after items 2, 5, 14 and 23 have been excluded and after response options *strongly disagree* have been recoded as 0, *moderately disagree* and *slightly disagree* as 1, *slightly agree* and *moderately agree* as 2, and *strongly agree* as 3.

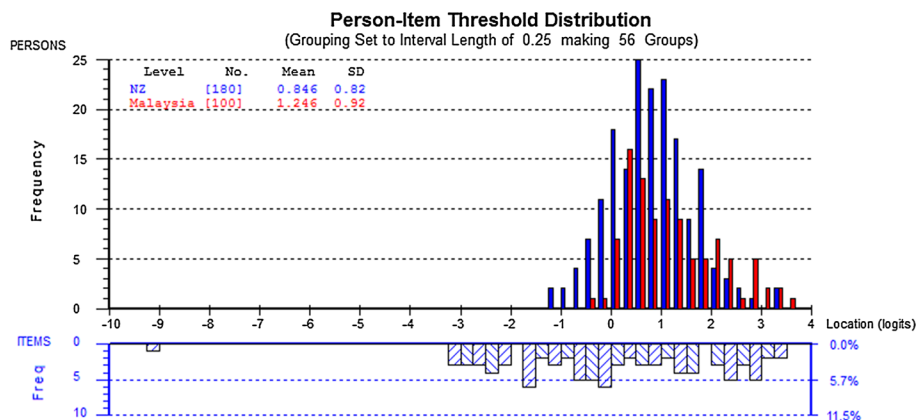


Fig. 3 Item-person threshold distribution after uniform rescaling of the OHQ items (Table 1, Analysis 2)

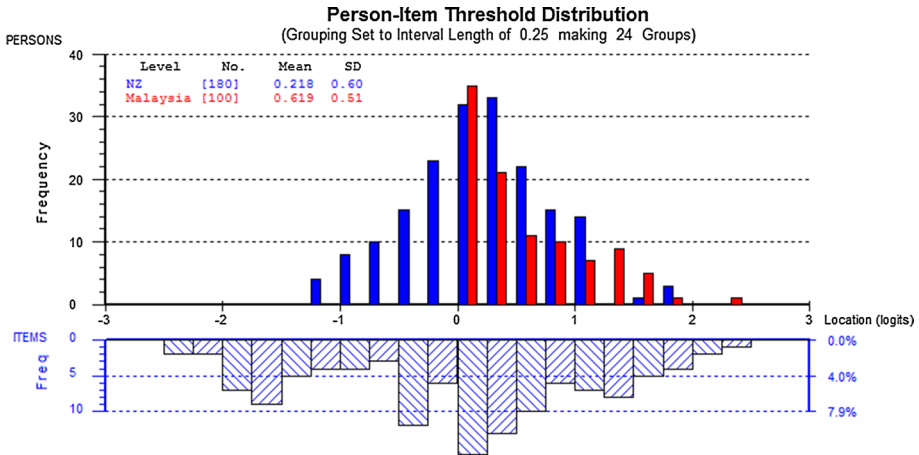


Fig. 4 Item-person threshold distribution for the final solution of the OHQ after removing misfitting items 2, 5, 14 and 23, creating three subtests and splitting items working differently for each sample (DIF) (Table 1, Analysis 4)

agree as 3. Then, the recoded 25 item responses (score range 0–75) should be added for each person.

Corresponding interval-level scores are presented on the right-hand side in both logit units and the raw score scale metric for convenience. Researchers who are currently using the OHQ or have already collected their data can use this algorithm to increase the precision of the OHQ. Considering the DIF by sample reported above, two different conversion tables are presented for use with samples comparable to the New Zealand and to the Malaysian student populations. It should be noted that the proposed ordinal-to-interval conversion algorithm can be used without the need to modify the original OHQ response format. However, these conversion tables can only be used for the respondents without missing data. The range of scores from 0 to 75 was used for practical reason because this range corresponds to the range of scores of the modified OHQ version and permits comparisons between ordinal and interval scores within the same range. For instance, ordinal score of 20 will correspond to the interval score of 21.93 (New Zealand sample) and 28.39 (Malaysian sample). This presentation was implemented to simplify comparison of ordinal and interval scores to satisfy a wider readership unfamiliar with Rasch analysis. However, the interval scores can be scaled to any possible range of scores (i.e. 1–100) and the authors may be contacted to assist with score conversion.

4 Discussion

The OHQ (Hills and Argyle 2002) is a widely-used ordinal measure of subjective happiness, but so far the precision of the instrument has not been investigated in sufficient detail. The current study used modern strategies of Rasch analysis to advance the psychometric properties and precision of the OHQ up to an interval-level measure. The results show that good fit to the Rasch model was achieved by minor modifications including uniform item rescaling, removing four misfitting items and combining locally dependent items into subtests. The psychometric properties of the modified 25-item OHQ are

Table 4 Converting from a uniformly rescored 25-item OHQ raw score (0–75) to an interval scale in logit units and in the same scale metric (0–75) for the New Zealand and Malaysian student populations

Raw score ^a	Interval measure				Raw score ^a	Interval measure			
	New Zealand		Malaysian			New Zealand		Malaysian	
	Logit	Scale	Logit	Scale		Logit	Scale	Logit	Scale
0	−3.37	0.00	−2.27	0.00	38	−0.42	33.99	0.16	31.55
1	−2.90	5.44	−2.03	3.19	39	−0.35	34.76	0.17	31.76
2	−2.62	8.72	−1.84	5.54	40	−0.28	35.54	0.19	31.94
3	−2.44	10.70	−1.71	7.29	41	−0.21	36.32	0.20	32.14
4	−2.32	12.12	−1.60	8.74	42	−0.15	37.11	0.22	32.37
5	−2.22	13.23	−1.49	10.11	43	−0.08	37.90	0.23	32.54
6	−2.14	14.14	−1.39	11.45	44	−0.01	38.68	0.25	32.76
7	−2.07	14.94	−1.28	12.92	45	0.06	39.46	0.27	33.00
8	−2.01	15.64	−1.15	14.56	46	0.13	40.24	0.29	33.26
9	−1.96	16.26	−0.93	17.40	47	0.20	41.02	0.31	33.54
10	−1.91	16.84	−0.41	24.15	48	0.26	41.79	0.33	33.88
11	−1.86	17.40	−0.44	23.86	49	0.33	42.56	0.37	34.29
12	−1.82	17.91	−0.23	26.58	50	0.40	43.32	0.41	34.83
13	−1.77	18.42	−0.19	27.09	51	0.46	44.07	0.46	35.50
14	−1.73	18.91	−0.17	27.36	52	0.53	44.82	0.52	36.24
15	−1.69	19.41	−0.15	27.59	53	0.59	45.55	0.58	37.02
16	−1.64	19.90	−0.14	27.78	54	0.65	46.28	0.64	37.83
17	−1.60	20.40	−0.12	28.00	55	0.71	46.99	0.70	38.63
18	−1.56	20.89	−0.11	28.16	56	0.78	47.70	0.76	39.45
19	−1.51	21.41	−0.10	28.24	57	0.84	48.42	0.83	40.27
20	−1.47	21.93	−0.09	28.39	58	0.90	49.12	0.89	41.12
21	−1.42	22.46	−0.07	28.59	59	0.96	49.82	0.96	41.98
22	−1.37	23.00	−0.06	28.72	60	1.02	50.54	1.02	42.85
23	−1.32	23.56	−0.06	28.80	61	1.09	51.26	1.09	43.75
24	−1.27	24.15	−0.04	28.98	62	1.15	52.00	1.16	44.66
25	−1.22	24.74	−0.03	29.11	63	1.22	52.76	1.24	45.61
26	−1.17	25.36	−0.03	29.20	64	1.29	53.56	1.31	46.58
27	−1.11	26.00	−0.01	29.39	65	1.36	54.38	1.39	47.62
28	−1.05	26.66	0.00	29.50	66	1.43	55.26	1.47	48.72
29	−1.00	27.33	0.01	29.68	67	1.52	56.20	1.56	49.87
30	−0.94	28.03	0.02	29.85	68	1.60	57.21	1.66	51.15
31	−0.87	28.73	0.04	30.03	69	1.70	58.33	1.77	52.57
32	−0.81	29.46	0.05	30.23	70	1.81	59.58	1.90	54.21
33	−0.75	30.19	0.07	30.43	71	1.94	61.04	2.05	56.15
34	−0.68	30.93	0.09	30.67	72	2.09	62.85	2.23	58.57
35	−0.62	31.69	0.10	30.89	73	2.30	65.26	2.48	61.83
36	−0.55	32.45	0.12	31.12	74	2.63	69.04	2.88	66.96
37	−0.48	33.22	0.14	31.35	75	3.15	75.00	3.49	75.00

OHQ, Oxford Happiness Questionnaire

^a To calculate the 25-item OHQ raw score exclude items 2, 5, 14 and 23; uniformly rescore other 25 OHQ items as following: 1 to 0; 2 to 1; 3 to 1; 4 to 2; 5 to 2; 6 to 3; and add rescored values together for each person. This table cannot be used for respondents with missing data

therefore robust, and ordinal responses to the original OHQ items can be transformed into an interval-level scale using the conversion algorithms provided in Table 4. This transformation can be conducted without the need to modify the original 29-item OHQ response format meaning that existing datasets can easily be re-analyzed to provide interval-level measurement.

Similar to Medvedev et al. (2016a), uniform rescaling has proven to be the best strategy to improve disordered thresholds of all OHQ items, as illustrated in Fig. 1 (disordered thresholds) and Fig. 2 (ordered thresholds after rescaling). These findings provide support for the four options response format used in the OHI (Argyle et al. 1989) as the most appropriate to use with the OHQ items. However, the OHI uses four statements as response options, which increases both questionnaire length and completion time. Alternatively, four Likert-scale response options defined as *strongly disagree* = 1, *slightly disagree* = 2, *slightly agree* = 3, *strongly agree* = 4, can be used because distinctions between *agree* and *slightly agree*, and between *slightly disagree* and *disagree* were shown to be unreliable (Fig. 1).

Items 2, 5, 14 and 23 critically affected individual estimates and appeared to be poorly related to the latent trait of happiness based on both Rasch and conventional results. Item 2 'I am intensely interested in other people' may have a face validity issue because it implies that happy people should be extremely preoccupied with other people, which is not supported by our results. It seems more likely that happy people naturally attract others and become an object of interest. Item 5 'I rarely wake up feeling rested' seems to focus on perceived level of physical energy at the time point of waking up and does not appear to be a reliable estimate of the latent trait according to our data. Perceived energy levels may vary substantially during the day, and assessing levels after waking up only is unlikely to reflect overall happiness. The results show that item 14 'There is a gap between what I would like to do and what I have done' is also not a reliable estimate of individual happiness levels. This item seems not to be fully consistent with the theory emphasising a cognitive component of happiness (Diener 1984; Diener et al. 2005) because an individual might be totally happy with the state of affairs regardless of a gap. This item might need to be reworded and ask about satisfaction with actual achievements rather than about the difference between one's plans and facts. Finally, item 23 'I don't find it easy to make decisions' implicitly suggests that lack of difficulty when making decisions contributes to greater happiness, which is not supported by our results and available theories on well-being and happiness. Modern life is challenging, and making a decision that ensures future well-being may involve thorough investigation of a subject matter, which is not necessarily an easy task. These four misfitting items were removed, which improved precision of the proposed 25-item OHQ version.

An alternative method to analyse the present data could have been Mokken analysis (Stewart et al. 2010). However, Mokken analysis only confirms that a scale meets the criteria for a sound and reliable ordinal scale—the items are ordered in a hierarchy of difficulty but the distances between items are not necessarily equal. One advantage of Rasch over Mokken analysis is that it permits the identification of a true interval scale. The disparity between findings from the Mokken analysis (Stewart et al. 2010) and the current Rasch analysis are due to differences between these two methods in dealing with misfitting items. In particular, if an item doesn't meet the expectations of the Mokken analysis (e.g. item is very similar to another item), the only available option for a researcher is to exclude that item. However, if an item doesn't fit expectations of a Rasch model for the same reason, which refers to local dependency in Rasch terminology, locally dependent items can be combined into subtests without a need to remove any of them. This is the second

advantage of Rasch modelling, namely that it permits fine tuning of a scale without a need to remove a large amount items. The subtests approach based on Lundgren-Nilsson et al. (2013) was effective in solving local dependency issues found between items of three distinct groups. Interestingly, one group of locally dependent items included all negatively worded items of the OHQ loaded together on one factor, and other items were split into more cognitive and more affective clusters (Table 3) confirming theoretical expectations (Diener 1984; Diener et al. 2005). However, after locally dependent items were combined into three subtests, unidimensionality of the 25-item OHQ was clearly evident, confirming structural validity of the modified scale. Our results are consistent with earlier research (Argyle 2001; Hills and Argyle 1998, 2002; Joseph and Lewis 1998) suggesting single dimension of happiness, which contains positive and negative affective components together with cognitive facets such as life satisfaction and happy traits.

One group of items works differently for Malaysian students compared to New Zealand as evidenced by DIF found between the New Zealand and Malaysian samples, and as a result separate ordinal-to-interval conversion tables were generated for each sample (Table 4). Figure 3 shows the item-person threshold distribution after uniform rescoring but before other Rasch modifications such as removing misfitting items, creating subtests and DIF split. There are items located outside the sample range of happiness trait, gaps uncovered by items and signs of a ceiling effect, which increase measurement error. Figure 4 illustrates the item-person threshold distribution of the 25-item OHQ after Rasch modifications including removal of misfitting items 2, 5, 14 and 23, creating subtests and splitting by sample DIF. It can be seen that the range of happiness levels possessed by the sample are perfectly covered by the item thresholds of the modified OHQ version, which is clear evidence for successful completion of this Rasch analysis.

The limitations of this study have to be acknowledged. The student samples may not reflect the full diversity of New Zealand's or the Malaysian's ethnicities, and no attempts were made to sample under-represented groups. Also, these findings were not replicated with a sample derived from the general population, which could increase generalizability.

The important contribution of this study is to allow researchers and clinicians to convert the OHQ raw score from an ordinal- to an interval-level scale and to use transformed data with parametric statistics without violating their fundamental assumptions. Also, the interval-level scores of the latent happiness factor provide researchers with the opportunity to study the effects of moderators and mediators of happiness in various contexts. OHQ scores that are transformed to an interval scale can be analysed reliably in such models because local dependency issues and potential item biases (DIF) are ultimately addressed by Rasch analysis if the data fits the model. The improved precision of the instrument may also be useful in clinical assessment, where the subjective wellbeing score might reflect recovery from psychological conditions such as anxiety, stress and depression.

5 Conclusion

Striving to increase subjective wellbeing or happiness in society with modern psychological approaches requires accurate assessment of the construct to determine its reliable predictors and obscuring factors. The current study conducted Rasch analysis of the OHQ and has demonstrated that after minor modifications the 25-item OHQ version

satisfies the Rasch model expectations. Therefore, the OHQ's precision can be increased up to an interval-level scale by using the ordinal-to-interval conversion tables presented here. These findings permit researchers and clinicians to use the OHQ data with parametric statistics and to make valid comparisons with interval- and ratio-level data such as electroencephalogram, heart rate or blood sugar levels.

Acknowledgments This study is a part of doctoral work of the first author funded by the Vice-Chancellor's Scholarship of the Auckland University of Technology.

Compliance with Ethical Standards

Conflict of interest We declare hereby that we have no conflicts of interest in regard to this paper.

Ethics Approval The study was conducted in compliance with the authors' institutional ethics committees.

Informed Consent Informed consent was received from all participants.

References

- Allen, M. J., & Yen, W. M. (1979). *Introduction on to measurement theory*. Monterey, CA: Brooks/Cole.
- Andrews, F. M., & McKennell, A. C. (1980). Measures of self-reported well-being: Their affective, cognitive, and other components. *Social Indicators Research*, 8(2), 127–155.
- Andrich, D., & Hagquist, C. (2013). Real and artificial differential item functioning. *Journal of Educational and Behavioural Statistics*, 37(3), 387–416.
- Andrich, D., Sheridan, B., & Luo, G. (2009). *RUMM 2030*. Perth: RUMM Laboratory.
- Argyle, M. (2001). *The psychology of happiness* (2nd ed.). New York: Taylor & Francis Inc.
- Argyle, M., Martin, M., & Crossland, J. (1989). Happiness as a function of personality and social encounters. In J. P. Forgas & J. M. Innes (Eds.), *Recent advances in social psychology: An international perspective*. Amsterdam: Elsevier.
- Beck, A. T. (1991). Cognitive therapy: A 30-year retrospective. *American Psychologist*, 46(4), 368–375.
- Beck, A. T., Steer, R. A., & Brown, G. K. (1996). *Manual for the Beck Depression Inventory II*. San Antonio: Psychological Corporation.
- Brandburn, N. M. (1969). *The structure of psychological well-being*. Chicago: Aldine.
- Carr, A. (2004). *Positive psychology*. New York: Brunner-Routledge.
- Compton, W. C. (2005). *Introduction to positive psychology*. Belmont: Wadsworth.
- Csikszentmihalyi, M. (1992). *The psychology of happiness*. Kent: Mackays of Chatham plc.
- Diener, E. (1984). Subjective well-being. *Psychological Bulletin*, 95(3), 542–575.
- Diener, E. (2006). Guidelines for national indicators of subjective well-being and ill-being. *Journal of Happiness Studies*, 7(4), 397–404.
- Diener, E., Lucas, R. E., & Oishi, S. (Eds.). (2005). *Subjective well-being: The science of happiness and life satisfaction*. New York: Oxford University Press.
- Diener, E., Sapyta, J. J., & Suh, E. M. (1998). Subjective well-being is essential to well-being. *Psychological Inquiry*, 9(1), 33–37.
- Eid, M., & Larsen, R. J. (Eds.). (2008). *The science of subjective well-being*. New York: The Guilford Press.
- Ellis, A. (2002). *Overcoming resistance: A rational emotive behavior therapy integrated approach* (2nd ed.). New York: Springer Publishing Company Inc.
- Fordyce, M. W. (1988). A review of research on the happiness measures: A sixty second index of happiness and mental health. *Social Indicators Research*, 20(4), 355–381.
- Fox, C. M., & Jones, J. A. (1998). Uses of Rasch modeling in counseling psychology research. *Journal of Counseling Psychology*, 45(1), 30–45.
- Hills, P., & Argyle, M. (1998). Positive moods derived from leisure and their relationship to happiness and personality. *Personality and Individual Differences*, 25(3), 523–535.
- Hills, P., & Argyle, M. (2002). The Oxford Happiness Questionnaire: A compact scale for the measurement of psychological well-being. *Personality and Individual Differences*, 33(7), 1073–1082.
- Joseph, S., & Lewis, C. A. (1998). The depression-happiness scale: Reliability and validity of a bipolar self-reported scale. *Journal of Clinical Psychology*, 54(4), 537–544.

- Joshanloo, M. (2015). Revisiting the empirical distinction between hedonic and eudaimonic aspects of well-being using exploratory structural equation modeling. *Journal of Happiness Studies*,. doi:[10.1007/s10902-015-9683-z](https://doi.org/10.1007/s10902-015-9683-z).
- Kahneman, D., & Krueger, A. B. (2006). Developments in the measurement of subjective well-being. *Journal of Economic Perspectives*, 20(1), 3–24.
- Kashdan, T. B. (2004). The assessment of subjective well-being (issues raised by the Oxford Happiness Questionnaire). *Personality and Individual Differences*, 36(5), 1225–1232.
- Kim-Prieto, C., Diener, E., Tamir, M., Scollon, C. N., & Diener, M. (2005). Integrating the diverse definitions of happiness: A time-sequential framework of subjective well-being. *Journal of Happiness Studies*, 6(3), 261–300.
- Kristoffersen, I. (2010). The metrics of subjective wellbeing: Cardinality, neutrality and additivity. *The Economic Record*, 86(272), 98–123.
- LeDoux, J. E. (2000). Emotion circuits in the brain. *Annual Review of Neuroscience*, 23, 155–184.
- Linacre, J. M. (1994). Sample size and item calibration stability. *Rasch Measurement Transactions*, 7, 328.
- Lundgren-Nilsson, Å., Jonsdottir, I. H., Ahlborg, G., & Tennant, A. (2013). Construct validity of the Psychological General Well Being Index (PGWBI) in a sample of patients undergoing treatment for stress-related exhaustion: A Rasch analysis. *Health and Quality of Life Outcomes*, 11, 2.
- Marais, I., & Andrich, D. (2008). Effects of varying magnitude and patterns of response dependence in the unidimensional Rasch model. *Journal of Applied Measurement*, 9(2), 105–124.
- Mayers, D. G. (1992). The secrets of happiness. *Psychology Today*, 25(4), 38–45.
- Medvedev, O. N., Shepherd, D., & Hautus, M. J. (2015). The restorative potential of soundscapes: A physiological investigation. *Applied Acoustics*, 96, 20–26.
- Medvedev, O. N., Siegert, R. J., Feng, X. J., Billington, D. R., Jang, J. Y., & Krägeloh, C. U. (2016a). Measuring trait mindfulness: How to improve the precision of the Mindful Attention Awareness Scale using a rasch model. *Mindfulness*, 7, 384–395.
- Medvedev, O. N., Siegert, R. J., Kersten, P., & Krägeloh, C. U. (2016b). Rasch Analysis of the Kentucky Inventory of Mindfulness Skills. *Mindfulness*, 7, 466–478.
- Nesse, R. M. (1990). Evolutionary explanation of emotions. *Human Nature*, 1(3), 261–289.
- Norquist, J. M., Fitzpatrick, R., Dawson, J., & Jenkinson, C. (2004). Comparing alternative rasch-based methods vs raw scores in measuring change in health. *Medical Care*, 42(1), I25–I36.
- OECD. (2015). The Organisation for Economic Co-operation and Development (OECD) Better life Index. <http://www.oecdbetterlifeindex.org/>.
- Pomplun, M., & Custer, M. (2005). The construct validity of the Stanford-Binet 5 measures of working memory. *Assessment*, 12(3), 338–346.
- Rasch, G. (1960). *Probabilistic models for some intelligence and attainment test*. Copenhagen: Danish Institute for Educational Research.
- Rasch, G. (1961). On general laws and the meaning of measurement in psychology. University of California Press. *Symposium conducted at the meeting of the Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability, Berkeley, California*.
- Robbins, M., Francis, L. J., & Edwards, B. (2010). Happiness as stable extraversion: Internal consistency reliability and construct validity of the Oxford Happiness Questionnaire among undergraduate university students. *Current Psychology*, 29(2), 89–94.
- Siegert, R. J., Tennant, A., & Turner-Stokes, L. (2010). Rasch analysis of the Beck Depression Inventory-II in a neurological rehabilitation sample. *Disability and Rehabilitation*, 32(1), 8–17.
- Skevington, S. M., MacArthur, P., & Somerset, M. (1997). Developing items for the WHOQOL: An investigation of contemporary beliefs about quality of life related to health in Britain. *British Journal of Health Psychology*, 2(1), 55–72.
- Smith, E. V. (2002). Detecting and evaluation the impact of multidimensionality using item fit statistics and principal component analysis of residuals. *Journal of Applied Measurement*, 3, 205–231.
- Spruk, R., & Kešeljević, A. (2015). Institutional origins of subjective well-being: Estimating the effects of economic freedom on national happiness. *Journal of Happiness Studies*,. doi:[10.1007/s10902-015-9616-x](https://doi.org/10.1007/s10902-015-9616-x).
- Stewart, M. E., Watson, R., Clark, A., Ebmeier, K. P., & Deary, I. J. (2010). A hierarchy of happiness? Mokken scaling analysis of the Oxford Happiness Inventory. *Personality and Individual Differences*, 48(7), 845–848.
- Stucki, G., Daltroy, L., Katz, J. N., Johannesson, M., & Liang, M. H. (1996). Interpretation of change scores in ordinal clinical scales and health status measures: The whole may not equal the sum of the parts. *Journal of Clinical Epidemiology*, 49(7), 711–717.

- Tellegen, A., Lykken, D. T., Bouchard, T. J., Wilcox, K. J., Segal, N. L., & Rich, S. (1988). Personality similarity in twins reared apart and together. *Journal of Personality and Social Psychology*, 54(6), 1031–1039.
- Tennant, A., & Conaghan, P. G. (2007). The Rasch measurement model in rheumatology: What is it and why use it? When should it be applied, and what should one look for in a Rasch paper? *Arthritis and Rheumatism*, 57(8), 1358–1362.
- Wainer, H., & Kiely, G. (1987). Item clusters and computerized adaptive testing: A case for testlets. *Journal of Educational Measurement*, 24(3), 185–201.
- Wilson, M. (2005). *Constructing measures*. Mahwah, NJ: LEA.
- Wright, B. D., & Stone, M. H. (1979). *Best test design*. Chicago: MESA Press.